RAMTIN SOLEYMANI

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EDUCATION

Toronto Metropolitan University **Toronto, ON**
Bachelor of Engineering, Specialization in Electrical Engineering (Co-op) *June 2026*

- Relevant Coursework: Digital Systems, SoC Design, Computer Architecture, Embedded Systems, Low-Power Digital ICs, Control Systems, Electronic Circuits I & II

PROJECTS

FPGA Camera/Display Pipeline
SystemVerilog, UVM, CDC

- Designed and verified a 320x240 OV7670-to-VGA SystemVerilog pipeline with SCCB sensor control, scaling output to 640x480.
- Resolved a 2-clock-domain metastability risk by implementing an asynchronous FIFO with Gray-coded CDC pointers, ensuring stable pixel transfer.
- Reduced frame-tear severity by 32.91% across 25 test frames by adding FIFO buffering, validating both architectures with UVM scoreboards for functional correctness.

Network-on-Chip Interconnect
SystemC, Self-Checking Verification

- Modelled a 4x4 Network-on-Chip with 16 wormhole-routed nodes, supporting both mesh and torus topologies.
- Verified zero routing errors or deadlocks across 384 packets and 1,920 flits using automated self-checking testbenches.
- Improved uniform-traffic latency by 1.9% (25.62 ns → 25.14 ns) by implementing torus wraparound links, demonstrating interconnect performance tuning.

32-bit Multi-Cycle CPU
RTL Design, STA, Timing Closure

- Architected a 32-bit SystemVerilog CPU with Harvard architecture and 19-state FSM control, deployed on Basys 3 and DE2-115 FPGAs.
- Closed timing at 100 MHz by restructuring the FSM by one cycle, improving Worst Negative Slack (WNS) from -0.742 ns to +3.275 ns.
- Reduced resource utilization by ~30% through datapath reuse; functionally verified 30+ instructions using Quartus II waveform simulation.

WORK EXPERIENCE

Hydro One Networks Inc **Toronto, ON**
Electrical Engineering Intern *2024 - 2025*

- Root-caused recurring 115 kV and 230 kV transmission faults by correlating outage history against weather data, isolating a lightning-backflash failure mode and validating a targeted grounding fix alongside a P.Eng.
- Modelled and verified 500 kV line safety margins in PLS-CADD, analyzing 60 Hz EMF and induced-current simulation output against IEEE and internal standards to confirm design compliance ahead of construction.
- Built a Python-integrated Power Automate script to auto-generate Excel status reports, cutting manual reporting effort by ~50% and eliminating a recurring source of tracking error.

R.J Burnside & Associates Limited **Toronto, ON**
Electrical Design Intern *2023 - 2024*

- Delivered 10+ electrical design drawings in AutoCAD while carrying a full course load, meeting all client and regulatory deadlines.
- Reduced projected costs up to 40% across 5+ street-lighting designs through systematic layout and equipment optimization.

TECHNICAL SKILLS

HDL/Verification:	SystemVerilog, Verilog, UVM, SVA, Functional Coverage, Waveform Debug, Scoreboards,
EDA Tools:	Synopsys VCS, ModelSim, Xilinx Vivado, Intel Quartus II, Multisim/SPICE
Digital Design:	RTL Design, CDC, STA, Synthesis, Timing Closure, Memory Hierarchy, Cache Coherence
Interfaces:	AXI/AHB, UART, I2C, ARM Cortex-M3, GPU Architecture
Scripting:	Python, Tcl, C/C++, Git